# ORF 245: Fundamentals of Statistics – Spring 2025

## Precept 3

1. Some examples for the discrete random variable:

   1. **Bernoulli Distribution**

   $$P(X = x) = p^x(1-p)^{1-x}, \quad x \in \{0, 1\}$$

   2. **Binomial Distribution**

   $$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, 2, \ldots, n$$

   3. **Geometric Distribution**

   $$P(X = k) = (1-p)^{k-1} p, \quad k = 1, 2, 3, \ldots$$

   4. **Poisson Distribution**

   $$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \ldots$$

2. (**Review of continuous random variable**)

   - Definition of CDF $F(x) = \mathbb{P}(X \le x)$ and PDF $f(x)$. The PDF is defined as the function satisfying

   $$\forall -\infty \le a \le b \le \infty, \qquad \mathbb{P}(a \le X \le b) = \int_a^b f(x)dx.$$

   - Their properties:
     - CDF: continuous, nondecreasing, $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$.
     - PDF: $\int f(x)dx = 1$, $f(x) \ge 0$.
   - The relationship between CDF and PDF: $F(x) = \int_{-\infty}^{x} f(t)dt$; $f(x) = F'(x)$.
   - Expectation and Variance:
     - $\mathbb{E}[g(X)] = \int g(x)f(x)dx$.
     - $\text{var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$
   - Some examples for the continuous random variable: Uniform $f(x) = \frac{1}{b-a} 1\{a \le x \le b\}$; Exponential $f(x) = \lambda e^{-\lambda x} 1\{x \ge 0\}$; Standard Gaussian $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

3. (**An Illustration: Linear Transformation of A Random Variable**)

   - $X$ continuous random variable, CDF $F(x)$, PDF $f(x)$.
   - $Y = aX + b$ for some constant $a, b$. $(a \ne 0)$ $\mathbb{E}[Y] = a\mathbb{E}[X] + b$; $\text{var}(Y) = a^2 \text{var}(X)$.
   - CDF, PDF of $Y$?
   - Example of Gaussian random variable $X \sim \mathcal{N}(\mu, \sigma^2)$.
     - $Z \sim N(0, 1)$ standard Gaussian, $X = \sigma Z + \mu \sim \mathcal{N}(\mu, \sigma^2)$.
     - $X \sim \mathcal{N}(\mu, \sigma^2)$, $\frac{X-\mu}{\sigma} \sim \mathcal{N}(0, 1)$

4. (**Continuous Random Variable**) The lifetime of a system is modeled by a random variable $X$ with PDF

$$f(x) = \begin{cases} kxe^{-x/2} & x > 0, \\ 0 & otherwise. \end{cases}$$

Question:

(a) Find the corresponding CDF of random variable $X$.

(b) Compute the value $\mathbb{P}(X > 5)$.

(c) Compute the expected lifetime $\mathbb{E}[X]$.